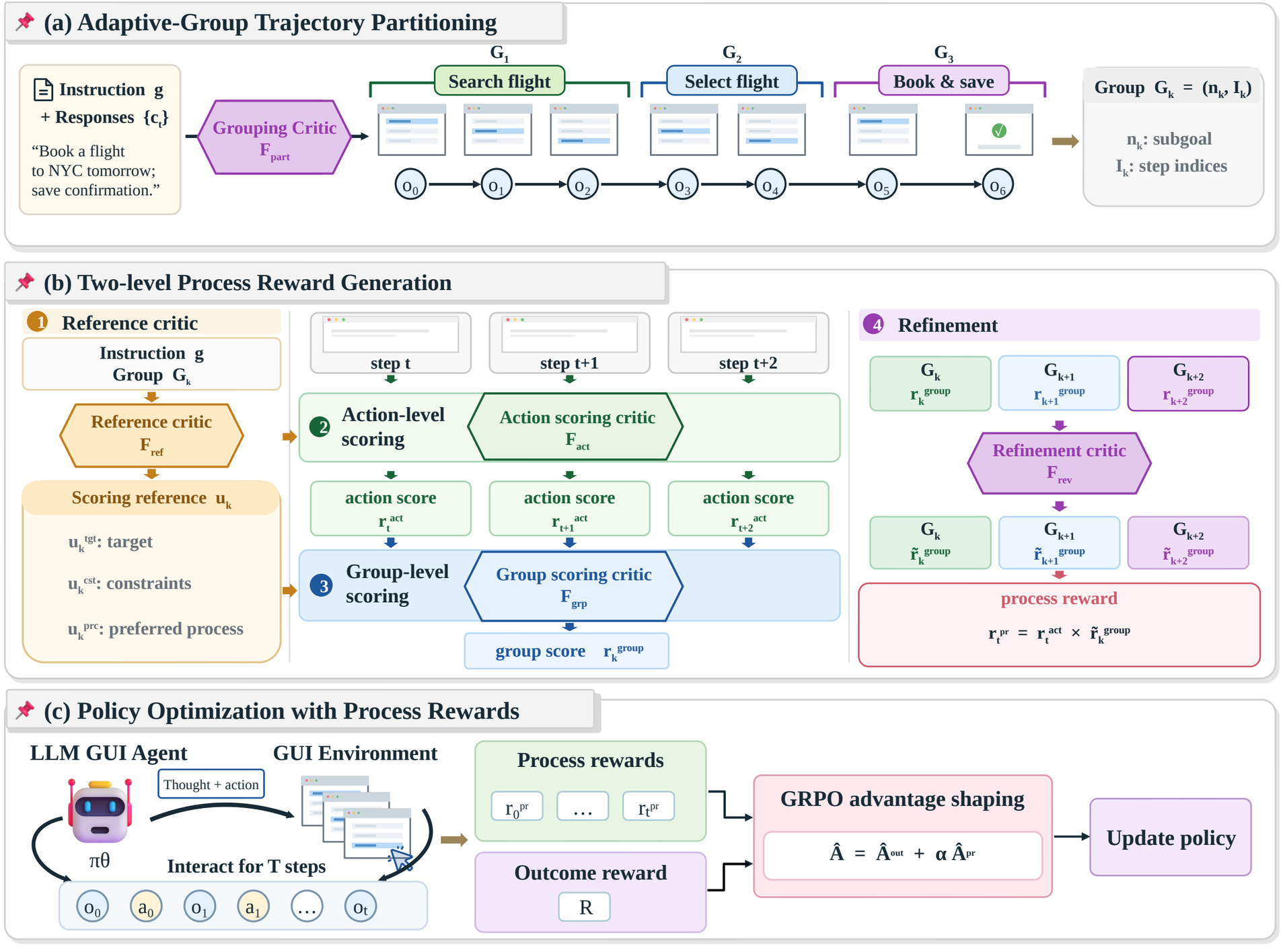

(a) Adaptive-Group Trajectory Partitioning
Instruction g
+ Responses {c_t}
"Book a flight to NYC tomorrow; save confirmation."
Grouping Critic F_part
G1
Search flight
G2
Select flight
G3
Book & save
o0 o1 o2 o3 o4 o5 o6
Group G_k = (n_k, I_k)
n_k: subgoal
I_k: step indices
(b) Two-level Process Reward Generation
1 Reference critic
Instruction g
Group G_k
Reference critic F_ref
Scoring reference u_k
u_k^tgt: target
u_k^cst: constraints
u_k^prc: preferred process
step t
step t+1
step t+2
2 Action-level scoring
Action scoring critic F_act
action score r_t^act
action score r_t+1^act
action score r_t+2^act
3 Group-level scoring
Group scoring critic F_grp
group score r_k^group
4 Refinement
G_k r_k^group
G_k+1 r_k+1^group
G_k+2 r_k+2^group
Refinement critic F_rev
G_k r̃_k^group
G_k+1 r̃_k+1^group
G_k+2 r̃_k+2^group
process reward
r_t^pr = r_t^act × r̃_k^group
(c) Policy Optimization with Process Rewards
LLM GUI Agent
GUI Environment
Thought + action
πθ
Interact for T steps
o0 a0 o1 a1 … ot
Process rewards
r_0^pr … r_t^pr
Outcome reward
R
GRPO advantage shaping
Â = Â^out + α Â^pr
Update policy